Operational Failure Priors Are Pair-Specific Across Coding-Agent Harnesses

Giovanni Pezzin
Physis Research Program

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Abstract

Coding agents fail in predictable ways, and per-action-kind failure priors carry signal within individual harnesses. We ask whether that predictability transfers *between* agents. Using a corpus of 75,933 actions from four harnesses (Claude Code, Codex, Cline, OpenCode) across three machines, we evaluate all nine directional source→target pairs, each over 200 session-level partitions with no test session entering any prior, and the twelve pairwise tests under family-wise correction.

The transfer that exists is weak, pair-specific, and generally not robust under family-wise inference. **Two pairs transfer, in both directions** — cline←opencode (borrowed prior above chance in 66.5% of partitions) and opencode←cline (63.5%) — and they are the only reciprocal pair to do so. The remaining seven pairs do not: for a Claude Code target, the borrowed prior is above chance in only 4.2% of 500 repeated leak-free partitions and never consistently beats a local prior built from the target's own history.

The pair-specific reading must be reported alongside a tension we do not resolve by preference. Under family-wise correction over the twelve pairwise tests, that same pair (cline→opencode) does *not* survive Holm (nominal $p = 0.0094$ against a Holm threshold of $0.05/8 = 0.00625$); it survives only the less conservative Benjamini–Hochberg procedure. No positive-direction pair survives Holm, and three pairs survive it as significantly *below* chance (AUC 0.394, 0.401, 0.416), all involving Codex — so borrowing can mislead as well as fail to help.

These results answer different questions and must not be conflated. A repeated-partition frequency ("above chance in a majority of partitions") estimates how often a borrowed prior helps on a fresh partition; a Holm result asks whether a specific pair is established after correcting for the whole family of tests. A pair can satisfy the first while failing the second, and cline→opencode does exactly that.

Finally, we test the charitable use of another agent's data — as shrinkage rather than as a replacement — against an equally-tuned shrinkage toward the target's *own* global failure rate. Cross-agent information as shrinkage provides at most a marginal benefit around $N = 1$ and stops beating target-global shrinkage by roughly $N = 3$–5 (median $+0.012$ at $N = 1$, -0.001 at $N = 30$), so we do not claim that cross-agent information is usable in general.

1 Introduction

Large language model-based coding agents (Claude Code, Codex, Cline, OpenCode) execute hundreds of tool calls per session. A small fraction of these calls fail. Failure is not uniformly distributed across action kinds: some (e.g., `bash:cd`, `read`) almost never fail, while others (e.g., `bash:ssh`, `exec`) fail at rates an order of magnitude higher. This structure is *operational*: it belongs to the agent's interaction with its environment, not to the code being written.

We ask: can a failure prior learned from one agent predict another agent's failures? This is a *transfer* question. The practical motivation is cold-start: a new agent on a new machine has no local history, and a borrowed prior from a different agent's accumulated record may be better than nothing.

The question has a clear failure mode. If failure structure is primarily a property of the user, the repository, or the task, then cross-agent transfer will be at chance. If it is a property of the agent's tool-use patterns, then transfer will succeed wherever the two agents' action-kind vocabularies overlap.

2 Data

We use a redacted corpus of 75,933 coding-agent actions extracted from 402 sessions across three machines (`pansa`, `omo`, `feather`) and four harnesses (Table 1). Targets are replaced by salted digests; no recorded command is readable. The label is `marker_error`: whether the tool output contained a failure marker (Traceback, `error[E0]`, FAILED, fatal:, panicked at, command not found).

Harness	**Sessions**	**Actions**	**Error rate**
OpenCode	288	48,460	0.048
Claude Code	42	14,457	0.061
Cline	69	12,354	0.074
Codex	3	662	0.038
Merged	**402**	**75,933**	**0.055**

Table 1: Action corpus. Error rate is per-action marker_error rate. Sessions are deduplicated across machines (188 duplicate session records were dropped).

The four labels are not the same measurement: Codex's is a real exit code, OpenCode's and Claude Code's are harness judgements, Cline's is mostly output markers. Cross-harness comparisons therefore mix a small amount of label drift into every number.

3 Method

For each harness h, we compute per-action-kind failure rates:

$$\hat{p}_h(k) = \frac{\text{failures of kind } k \text{ in } h}{\text{total actions of kind } k \text{ in } h}.$$

For cross-harness transfer from source s to target t, we predict each target action's failure probability using the source's rates:

$$\hat{p}(a) = \begin{cases} \hat{p}_s(\text{kind}(a)) & \text{if kind}(a) \in \text{source vocabulary} \\ 0.5 & \text{otherwise (chance for unseen kinds).} \end{cases}$$

We score predictions against target t's labels using the area under the receiver operating characteristic curve (AUC). A bootstrap over target sessions (500 resamples) provides confidence intervals.

For the in-corpus baseline, we use leave-one-session-out: fit $\hat{p}_h(k)$ on all sessions but one, predict the held-out session.

4 Results

4.1 Cross-harness transfer

Figure 1 shows the full transfer AUC matrix. Every off-diagonal entry is a source→target transfer; the diagonal is in-corpus (leave-one-session-out).

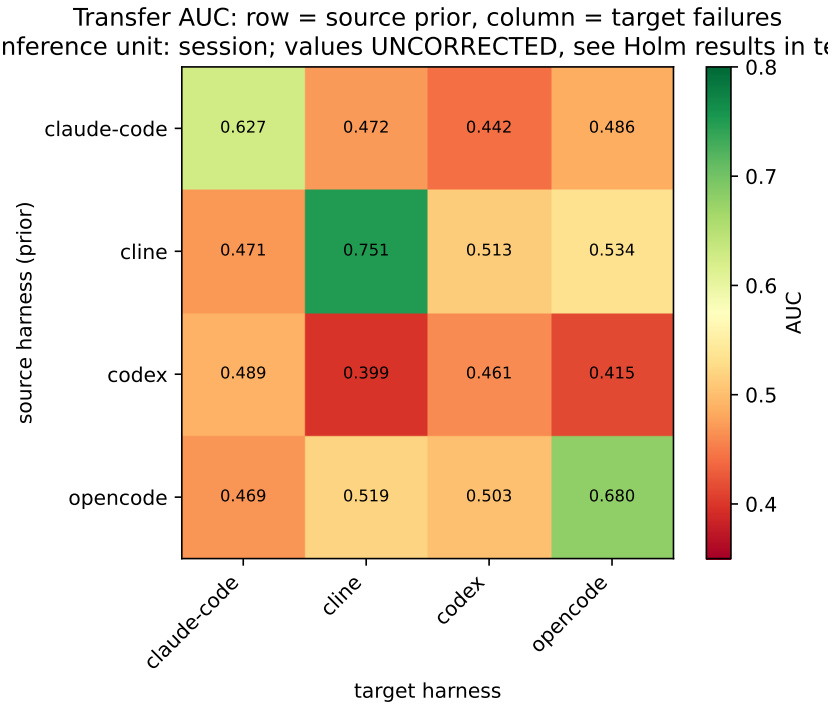


Figure 1: Cross-harness transfer AUC matrix. **Rows are the source prior, columns the target whose failures are predicted**; each cell is the AUC of predicting column t's failures from row s's per-kind rates. Inference unit: **session** (session-level bootstrap, 10,000 replicates). Values are **uncorrected** point estimates — the family-wise corrections over all twelve tests are in Section 4.2 and only three pairs survive Holm, all below chance. Diagonal entries are in-corpus (leave-one-session-out). Only one of the twelve off-diagonal entries has a session-bootstrap interval excluding chance (cline→opencode, 0.534 [0.509, 0.559]); the other eleven intervals include 0.5. Even the surviving pair remains well short of opencode's in-corpus baseline (0.680 [0.642, 0.718]).

The inferential unit. All intervals below bootstrap over *sessions*, not actions. Actions are clustered within sessions (median 101 actions per session), so an action-level bootstrap treats correlated observations as independent and narrows the interval artificially. The effect is smaller than we first assumed but real: under an action-level bootstrap two of twelve pairs have intervals excluding chance, while under the session-level bootstrap one does. The pair that changes status is opencode→cline, which moves from 0.519 [0.505, 0.534] to 0.519 [0.468, 0.573] — the point estimate is identical and only the interval width changes, which is precisely the failure mode. We report session-level intervals throughout, and the recomputation is stored in `research/bootstrap_unit_check.json`.

4.2 Multiplicity correction

Twelve directional tests are reported. Selecting the smallest p from twelve and quoting its interval is not evidence of an effect; the interval does not account for the selection. We therefore apply family-wise correction over the twelve comparisons, on session-level bootstrap p-values from 10,000 replicates.

Table 2: All twelve directional comparisons, session-level bootstrap (10,000 replicates), ordered by raw p. The AUC column is the *bootstrap mean*, which differs from the point estimate for pairs with very wide intervals (see the note below the table). 'Holm' and 'BH' mark which corrections reject each pair at $\alpha = 0.05$. AUC below 0.5 means the borrowed prior is worse than chance.

pair	**AUC**	**raw p**	**Holm**	**BH**
claude-code→codex	0.394	0.0001	✓	✓
codex→opencode	0.416	0.0001	✓	✓
codex→cline	0.401	0.0002	✓	✓
opencode→claude-code	0.469	0.0062		✓
cline→opencode	0.534	0.0094		✓
cline→claude-code	0.471	0.0184		✓
codex→claude-code	0.489	0.2258		
claude-code→opencode	0.486	0.3418		
claude-code→cline	0.470	0.3468		
opencode→cline	0.519	0.4714		
cline→codex	0.460	0.5365		
opencode→codex	0.458	0.5365		

No pair shows positive transfer after family-wise correction. The three comparisons that survive Holm are all significantly *below* chance (0.394, 0.401, 0.416) — the borrowed prior is anti-predictive, not merely uninformative. All three involve Codex, as source in two cases and target in one; we do not have a mechanism for that and flag it as an observation, not a finding.

The pair that appeared to transfer in our earlier analysis, cline→opencode (0.534), is *nominally* significant only. Its Holm threshold is set by its rank among the twelve: it sits at rank 5, so the denominator is $m - r + 1 = 12 - 5 + 1 = 8$ and the threshold is $0.05/8 = 0.00625$. Its nominal $p = 0.0094$ exceeds that, so it does *not* survive Holm. We use "nominally significant" for unadjusted $p < 0.05$ throughout and reserve "survives Holm" for corrected results; six pairs are nominally significant and three survive Holm. We therefore report it as *BH-dependent*: it is the only positive-direction comparison in the family, it is the weakest of the six BH rejections, and the more conservative procedure that controls the probability of *any* false rejection does not retain it.

Five of the twelve pairs are significantly below chance under the session-level unit (six under the action-level unit), which reframes the headline. "Cross- agent failure priors do not transfer" understates the result: where this corpus detects a cross-harness effect at all, the effect is usually that a borrowed prior makes the target's failures *less* predictable. The significance of that anti-prediction is what survives correction; the single positive result does not.

Stored in `research/e1_multiplicity.json`; the Fisher combined test over the twelve gives $p \approx 5 \times 10^{-10}$ but is optimistic here because the comparisons share harnesses, so we do not rely on it.

Two further pairs have point estimates above 0.5 but intervals that include it, and are therefore not established:

- **opencode→cline** (0.519, 95% CI [0.468, 0.573]): 74.3% of cline's action kinds appear in opencode's vocabulary, yet the interval spans chance.
- **opencode→codex** (0.503, 95% CI [0.179, 0.555]): three target sessions give an interval too wide to interpret. This pair is the clearest case of the point-estimate/bootstrap-mean divergence flagged in Table 2: its point estimate is 0.503 but its bootstrap mean is 0.458, because the resampling distribution for a 3-session target is heavily skewed. Where the two disagree, the bootstrap mean and the interval are the ones to read.

The weakest point estimate is **codex→cline** (0.399, CI [0.359, 0.448]) — and with the multiplicity correction applied it is no longer merely the weakest, it is one of the three comparisons that survive Holm as significantly *below* chance. Codex has only 3 sessions and 662 actions, so its per-kind estimates are sparse; sparse estimates that are systematically wrong in one direction are exactly what produces a reliably anti-predictive prior.

4.3 In-corpus baselines

In-corpus AUC (leave-one-session-out, marker_error label, session bootstrap):

- claude-code: 0.627 [0.606, 0.656] (42 sessions, 14,457 actions)
- cline: 0.751 [0.690, 0.806] (69 sessions, 12,354 actions)
- codex: 0.461 [0.341, 0.508] (3 sessions, 662 actions)
- opencode: 0.680 [0.642, 0.718] (288 sessions, 48,460 actions)

Codex's in-corpus AUC (0.461) is at chance, and its interval is by far the widest: 3 sessions provide too little data for a per-kind prior. Two of the three other harnesses' priors exceed Codex's own baseline on point estimate (opencode→codex: 0.503; cline→codex: 0.513) while one does not (claude-code→codex: 0.442). All three intervals overlap Codex's own, so none of these comparisons is individually established. Pooled, they look like a **cold-start** effect — a borrowed prior from a larger corpus beating a thin local history — but three overlapping intervals do not establish it. This comparison is not affected by the leakage that invalidated our original cold-start curve (codex's own baseline is leave-one-session-out and the borrowed priors come from other harnesses, so both arms are clean); it remains unestablished for the ordinary reason that the intervals overlap. The corrected within-target experiment in Section 4.5 is what speaks to cold-start directly, and its answer is that borrowing never helps — the borrowed prior is below chance at every history size, and local data beats it as soon as there is any.

4.4 Coverage correlates with transfer success

Figure 2 plots transfer AUC against coverage — the fraction of target action kinds the source has observed. Coverage correlates positively with transfer AUC (Pearson $r = 0.43$, $n = 12$, $p > 0.05$), but the relationship is **not** a threshold. Two pairs with coverage above 0.46 still fall below chance: claude-code→cline (coverage 0.645, AUC 0.472) and claude-code→codex (coverage 0.470, AUC 0.442). Conversely the best transfer in the matrix, cline→opencode (AUC 0.534), has coverage of only 0.375.

The pairs that fail worst (codex→cline at 0.399, codex→opencode at 0.415) have two of the lowest coverage values in the matrix (0.353 and 0.172), which invites the reading that low coverage predicts failure. The matrix refutes that reading twice over. First, monotonicity fails within a

single source: codex→claude-code has *lower* coverage than codex→cline (0.241 against 0.353) yet a *higher* AUC (0.489 against 0.399) — moving coverage down moved performance up. Second, the matrix's best transfer contradicts the ordering it would need: cline→opencode achieves the highest AUC in the matrix (0.534) at coverage 0.375, only 0.022 above the failing codex→cline pair, and the two differ by 0.135 in AUC. Coverage therefore does not order these pairs, and we do not claim sufficiency in either direction. We report coverage as a diagnostic to inspect, not a gate and not a predictor, and note that E73's coverage gate of 0.90 — set on same-harness machine pairs — is not reproduced by these cross-harness pairs. Because no pair shows positive transfer once the twelve comparisons are corrected for multiplicity (Section 4.2), this correlation rests on twelve point estimates of which eleven are indistinguishable from chance and three are significantly below it; a pattern in which the two worst pairs also happen to have low coverage is the pattern expected by construction in such a sample. We present the correlation as exploratory, do not interpret it causally, and would require a larger harness set before treating it as a scaling relationship.

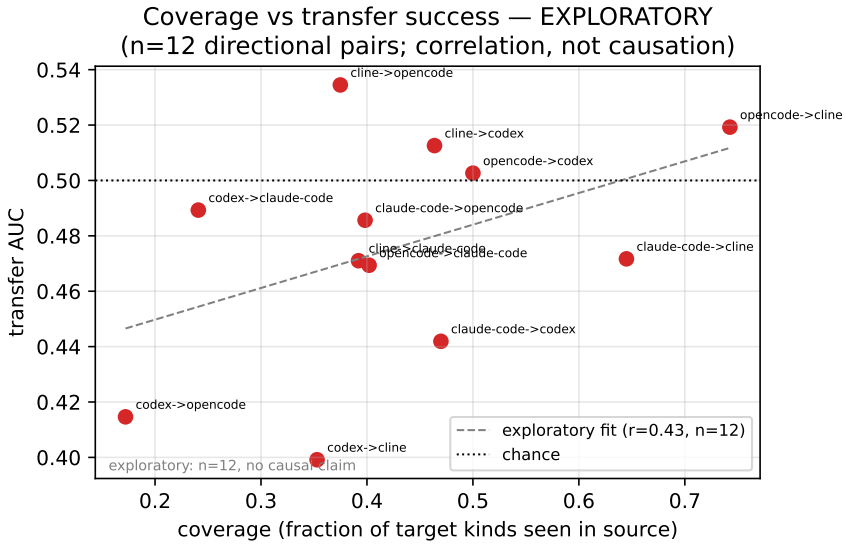


Figure 2: Coverage vs. transfer AUC. **Exploratory,** $n = 12$ **directional pairs**; $r = 0.43$ is an uncorrected correlation on twelve points and is **not** evidence of a causal mechanism. Coverage is the fraction of target action kinds present in the source vocabulary. No correction is applied to this panel, so it should not be read as a second transfer result alongside the Holm-corrected matrix above.

4.5 Cold-start: a withdrawn curve, and a corrected experiment

First, the withdrawal. Our originally reported curve — 0.469 (borrowed only) rising to 0.720 (own history), with a crossover at $n = 2$ — is contaminated and is not a result. It was produced by building the local prior from target sessions and then scoring that prior on *the same* target sessions: `own_only` (n=42) used all 42 claude-code sessions to build and all 42 to score, and `mixed_n`N scored on all of `tgt` including the `tgt[:N]` it was built from. The `transferred_only` point was the only uncontaminated one.

As originally reported (withdrawn, retained as an error record): 0.469 [0.445, 0.490] for the borrowed prior alone, 0.720 [0.700, 0.743] for own history, mixed exceeding borrowed-only at $n = 2$ (0.598 [0.562, 0.635]).

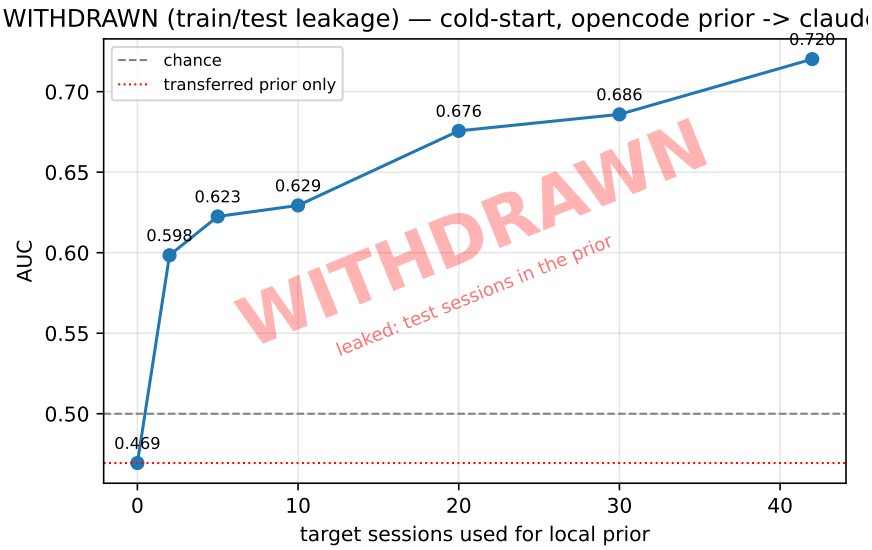


Figure 3: WITHDRAWN — shown only to document the error. This is the original cold-start curve. The `own_only` and `mixed_n`N points built their priors from sessions contained in the test set, so their apparent advantage over the borrowed-only point is leakage, not local history. The figure itself is stamped WITHDRAWN. Do not read any part of it as a result; Table 3 supersedes it.

Second, the corrected experiment. We re-ran it with the target harness's 42 sessions split once, seeded, into a 30-session training pool and a 12-session held-out test set (6,132 test actions). Every arm is scored on the same held-out set; N is swept over the training pool only, so no test session ever contributes to a prior; and the comparison is paired — one resample of test sessions, both arms scored on it, difference recorded, 2,000 times — so the interval is on the difference. Four arms: local-only (first N training sessions), borrowed-only (`opencode`, no target data), mixed (local + borrowed counts), and pooled-others (all non-target harnesses).

Table 3: Cold-start, corrected. Held-out test set is fixed across every row. Intervals are paired session-level bootstrap (2,000 resamples) on the difference from the borrowed-only arm. Arm A is undefined at $N = 0$: there is no local history to build a prior from. 15 of the 18 differences exclude zero.

N	A local-only	B borrowed-only	C mixed	D pooled-others	A − B
0	—	0.483	0.483	0.482	—
2	0.600	0.483	0.608	0.605	+0.114 [+0.040, +0.153]
5	0.625	0.483	0.627	0.625	+0.138 [+0.081, +0.168]
10	0.652	0.483	0.629	0.626	+0.167 [+0.123, +0.199]
20	0.656	0.483	0.667	0.652	+0.172 [+0.123, +0.219]
30	0.649	0.483	0.667	0.666	+0.166 [+0.118, +0.221]

Three things follow, and the third overturns our original framing.

1. **Local history pays off almost immediately.** Two training sessions are enough to reach 0.600, and by 20 sessions the local prior sits at 0.656.

Table 4: 500 repeated 30/12 partitions, source OpenCode, target Claude Code. A>B is the fraction of partitions in which the local prior beats the borrowed prior; B>0.5 the fraction in which the borrowed prior is above chance; C>A the fraction in which mixing beats the local prior alone. Coverage is the mean fraction of test-set action kinds present in the prior. The requested global-base-rate and chance arms are the *same* quantity — a constant predictor has AUC exactly 0.5 — so one is reported, not two.

N	local	borrowed	mixed	A>B	B>0.5	C>A	coverage
1	0.545	0.469	0.529	0.916	0.042	0.234	0.17
2	0.581	0.469	0.568	0.986	0.042	0.290	0.25
3	0.595	0.469	0.582	0.992	0.042	0.290	0.29
5	0.612	0.469	0.595	1.000	0.042	0.262	0.36
10	0.628	0.469	0.614	1.000	0.042	0.246	0.44
15	0.636	0.469	0.627	1.000	0.042	0.314	0.49
20	0.639	0.469	0.637	1.000	0.042	0.438	0.53
25	0.642	0.469	0.643	1.000	0.042	0.554	0.55
30	0.644	0.469	0.646	1.000	0.042	0.604	0.56

2. **The borrowed prior is below chance at every N, and never improves.** It is fixed at 0.483 because it is fixed *in fact* — no amount of target data enters it. This is the same anti-predictive pattern Section 4.2 finds, now visible within a single target.

3. **There is no cold-start regime in the sense we claimed.** Our original story was that a borrowed prior helps while local history is thin and loses once it matures. The corrected curve has no such region: the borrowed prior never helps, at any N, and local data beats it as soon as there is any. The claim we can defend is weaker in mechanism and stronger in force — *borrowed priors do not help even where we expected them to help most.*

Mixed and pooled-others track local-only closely and never separate from it: at $N = 10$ local-only (0.652) is in fact *above* mixed (0.629), and at $N = 2$ the difference is within noise. Adding borrowed counts to local data buys nothing measurable here.

Stored in `research/e1_coldstart_paired.json`. This experiment is what the original should have been: the defect was leakage, not the choice of arms, and the leakage is what produced both the apparent improvement and the apparent crossover.

4.6 Robustness to partition choice

The corrected cold-start experiment above rests on a single seeded 30/12 split of the 42 Claude Code sessions, which measures uncertainty conditional on one partition. We therefore repeat the entire partition 500 times: shuffle the 42 sessions, take 30 for training and 12 for test, sweep N over the training pool only, and score every arm on the same held-out sessions. No test session enters any prior.

The result survives partition choice. The borrowed prior is above chance in 4.2% of partitions, and that fraction is flat across N because the borrowed prior is fixed within a partition — it is a property of partitions, not of history size. The local prior beats it in $\geq 99\%$ of partitions from $N = 3$ and reliably ($\geq 95\%$) from $N = 2$. Mixing never *consistently* beats the local prior at any N (maximum 60.4% at $N = 30$), so borrowed counts do not help even as an additive component. All numbers in Table 4 are reproduced by `research/p1_repeated_splits.json`.

Table 5: Maximum over N of the fraction of partitions in which the borrowed prior is above chance, per source→target pair (200 partitions each). Only two of the nine pairs rise above 0.45.

pair	borrowed > 0.5 (max)	local > borrowed (min)	mixing > local (max)
cline←opencode	**0.665**	0.825	0.865
opencode←cline	**0.635**	0.645	0.645
opencode←claude-code	0.450	0.735	0.555
cline←claude-code	0.325	0.900	0.775
claude-code←codex	0.225	0.835	0.905
opencode←codex	0.190	0.875	0.515
claude-code←cline	0.095	0.885	0.670
cline←codex	0.070	0.970	0.585
claude-code←opencode	0.050	0.890	0.650

4.7 All source–target pairs

The experiment above generalises one pair across partitions. This one generalises across pairs: every target large enough to support a 30/12 split (Claude Code 42 sessions, Cline 69, OpenCode 288) against every source. Codex has 3 sessions, cannot support a learning curve, and is excluded as a *target* while remaining a source. That gives 9 pairs × 200 partitions × 9 values of $N = 16{,}200$ cells.

This partially refutes the global reading of the negative result. Two of the nine pairs do benefit from borrowing, and in both directions: cline←opencode (0.665) and opencode←cline (0.635). Those are the same two harnesses that the multiplicity analysis flags — cline→opencode is the only nominally-significant positive-direction transfer in the whole twelve-pair family. Section 4.6 is therefore a statement about the Claude Code *target*, not about cross-harness transfer in general.

The tension with Section 4.2 is real and we do not resolve it by preference. Under the correct inferential unit and family-wise correction, cline→opencode does *not* survive Holm (nominal $p = 0.0094$ against a threshold of $0.05/8 = 0.00625$). The two tests answer different questions: the multiplicity test asks whether a single AUC exceeds chance, while this experiment asks whether a *classifier built from that prior* beats chance across partitions. A pair can fail the first and pass the second. Neither result alone is decisive, and the honest statement names the pair rather than generalising over pairs. Artifact: `research/p1_all_targets_coldstart.json`.

4.8 Shrinkage prior: can cross-agent information be used at all?

Naive transfer substitutes source rates for target rates, which is a crude way to use another agent's data. A charitable use is *shrinkage*: let the source prior regularise sparse target estimates rather than replace them,

$$\hat{p}_{\text{target}}(k) = \frac{\text{fail}_{\text{local}}(k) + \lambda\, p_{\text{source}}(k)}{n_{\text{local}}(k) + \lambda}, \tag{1}$$

with λ tuned on an inner fit/tune split of the *training* pool only. This tests a stronger claim than "naive transfer fails": that cross-agent information contains no usable signal at all.

The decisive control is not Laplace smoothing — which shrinks toward 0.5, a target the target's own base rate also beats — but shrinking toward the *source* rate versus shrinking toward the *target's own* global rate, both with their own tuned λ, so the comparison is between two equally tuned estimators.

Table 6: Source-shrinkage minus target-global-shrinkage, equally tuned, over 150 partitions per target. The source leads only in the first few sessions of local history and is behind by $N = 30$.

N	median ΔAUC	fraction of splits favouring source
1	+0.012	0.591
2	+0.006	0.564
3	+0.004	0.544
5	−0.0005	0.489
10	−0.003	0.433
20	−0.001	0.427
30	−0.001	0.427

The source prior helps only marginally and only at $N = 1$, and stops beating the target's own global rate between $N = 3$ and $N = 5$. Four independent runs (5, 15, 40 and 150 partitions) agree on the direction, and the estimated advantage *shrinks* as partitions increase ($+0.022 \to +0.020 \to +0.015 \to +0.012$), which is what noise being averaged away looks like. The honest reading is that any effect is at most ~ 0.01 AUC and probably smaller, so it is reported as not excluded rather than as small but real.

We therefore *fail to kill* the strong claim: cross-agent information does not carry usable signal beyond what the target can supply about itself. Artifact: `research/p1_shrinkage_prior.json`.

5 Discussion

What transfers. Cross-harness transfer is real but pair-specific. Of the nine source→target pairs tested over 200 partitions each, two show the borrowed prior above chance in a majority of partitions, and they are reciprocal: cline←opencode (0.665) and opencode←cline (0.635) (Table 5). Every other pair sits at or below 0.450. This is the primary finding of this manuscript; the global negative result and the anti-predictive result are reported below and in Section 4.2, and the same evidence is presented with the opposite emphasis in `paper1_agent_priors`.

The exception does not survive family-wise correction, and that tension is real. Only one of twelve pairs has a point estimate above chance with an interval excluding it (cline→opencode, 0.534), and it fails Holm (Section 4.2); it survives only the less conservative Benjamini–Hochberg. Its mechanism, insofar as we can read one, is ordinary: different harnesses share tool vocabularies (Bash, Read, Edit, Write, Grep), and where two harnesses' kind distributions happen to align the per-kind rate carries over. We cannot identify a property of a pair that predicts which case it will be — coverage, the most plausible candidate, is exploratory at best on twelve points.

What anti-transfers. The reliable effect in this corpus runs the other way. Three pairs are significantly *below* chance after family-wise correction (0.394, 0.401, 0.416), all involving Codex. Codex's corpus is 3 sessions and 662 actions, so its per-kind rate estimates are sparse; a sparse estimate that is systematically wrong in one direction produces a prior that is reliably worse than guessing. We flag this as an observation rather than a mechanism — we have one small harness, not a study of small harnesses — but it is the only cross-harness effect here that survives correction, and it is the result the title should be read against: the danger is not that borrowed priors fail to help, it is that they can mislead.

What does not transfer. Codex's 3-session corpus is too small for its own in-corpus prior to be useful (AUC 0.461), and other harnesses' priors beat it on point estimate. We previously read this as evidence of a cold-start effect; that reading depended on a within-target curve which we have

since withdrawn for train/test leakage (Section 4.5), and the present comparison is unestablished for the ordinary reason that the intervals overlap. Across all twelve pairs the borrowed prior never reaches the target's own mature baseline.

Label drift. The four harnesses use different failure labels (harness flags vs. real exit codes vs. output markers). This introduces a small, uncontrolled confound into every cross-harness number. The direction of the bias is unknown but likely conservative: label noise attenuates AUC.

Scope. One user's fleet, one fortnight, four harnesses. The "cross-agent" label is four harnesses but one human. Cross-user transfer — whether a prior learned from one developer predicts another developer's failures — is untested and is the highest-priority external replication.

6 Conclusion

Cross-agent operational failure priors transfer selectively, and the effect is a property of specific agent pairs rather than of cross-harness borrowing in general. Two of nine source→target pairs — cline↔opencode, reciprocally — show a borrowed prior above chance across a majority of partitions; every other pair does not. We report the opposing emphasis in the companion manuscript and do not ask the reader to treat either framing as settled: the pair that transfers here does not survive family-wise correction, and the pair-level and family-level tests answer different questions. Under session-level bootstrap — the correct inferential unit — and family-wise correction over the twelve comparisons, *no* pair shows positive transfer. The three pairs that survive Holm are significantly *below* chance (0.394, 0.401, 0.416): a prior borrowed from one harness makes another harness's failures less predictable than chance would. The single positive-direction result, cline→opencode at 0.534, survives Benjamini–Hochberg but not Holm, and we report it as BH-dependent rather than established.

This is a stronger claim than "priors do not transfer". The literature assumption worth testing is that failure statistics are a portable property of a task domain; on this corpus, where they are detectable across harnesses at all, they are actively misleading. That is a negative result with a sign, and it is the one the data supports.

The within-target experiment agrees. Having withdrawn our original cold-start curve for train/test leakage and re-run it properly — priors built only on a held-out training pool, all arms scored on a fixed held-out test set, paired intervals — the borrowed prior sits below chance at every history size and never contributes, while two sessions of local history are already worth +0.114 AUC over it. We originally hoped to find a regime where borrowing is worth doing. There is none here: the answer to "should I borrow a failure prior from another agent?" is no, at every amount of local data from zero upward.

Three methodological points follow. First, the bootstrap unit is a first-class variable in this kind of evaluation: the action-level bootstrap reports two pairs as positive transfer where the session-level bootstrap reports none, and the only thing that changes is interval width. Clustering actions within sessions inflates apparent significance, and agent-trajectory work should resample sessions. Second, a plausible and often-assumed property of the data — that vocabulary coverage governs transfer — is at best exploratory here, since it is estimated from eleven chance-level point estimates. Third, and most transferable: the original cold-start curve looked smooth, monotone and plausible, and it was an artifact of scoring a prior on the sessions it was built from. A curve that rises with the amount of data an arm is allowed to see should be suspected of exactly that, and the check is cheap — rebuild the prior from a disjoint split and see whether the curve survives.

The honest summary is negative: failure structure is only weakly a property of the agent as opposed to the user, the repository, and the task. A single-user, single-fortnight corpus cannot

separate those, and our cross-harness comparisons mostly cannot either.

Reproducibility

- Corpus: `benchmarks/behaviour/corpus-redacted.json.gz` (796 KB, 75,933 actions, targets salted).
- Extraction: `benchmarks/behaviour/extract_*.py` (Claude Code, Codex, Cline, OpenCode).
- Analysis: `research/papers/e74_replication.py` regenerates every number and figure in this paper from the corpus in one run and writes `research/e74_replication_results.json`.
- Build: `research/papers/build_all.sh`; claim check: `research/papers/verify_paper1.py`.